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Editorial

Key technologies for electric vehicles

1. Introduction

Electric vehicles (EVs) are playing an increasingly important role in decarbonizing the transportation sector. They constitute a promising solution to a set of global challenges such as climate change and air pollution. EVs are an integration of a wide spectrum of techniques, such as battery monitoring, battery safety and vehicle energy management. In this regard, the EV development still faces significant challenges, which necessitate innovations in EV technologies. Given this, Green Energy and Intelligent Transportation (GEITS) organizes a special issue of "Key Technologies for Electric Vehicles" that attempts to advance knowledge in the area of EVs and provides a platform for researchers and engineers to share recent research results and discuss critical challenges in this field. A wide spectrum of topics are discussed, including but not limited to the following:

- Modeling and optimization of EVs
- Advanced battery design and management technology
- Automobile motors, power transmission system and control
- Human-machine collaborative interaction and intelligent decision-making
- Wire-controlled chassis, assisted driving technologies and internet of vehicles
- Advanced charging technologies
- Electrical and mechanical sensing technologies
- Infrastructure and auxiliary devices of EVs
- Full life cycle assessment, test evaluation, standards and regulations of EVs

2. Special issue content

The special issue on "Key Technologies for Electric Vehicles" reports on recent findings in the research and development of EVs. A total of 21 articles have been published, covering a variety of critical aspects of EVs.

He et al. [1] analyzed the development of battery EVs (BEVs) in China. The "technology system architecture for BEVs" is introduced, based on which critical technological points regarding BEV development are investigated in depth. Particularly, the progress and future opportunities of three core components, namely "BEV platform, charging/swapping station, and real-time operation monitoring platform" are systematically discussed. This study highlights the contributions made by China in developing BEVs and provides valuable sights into the future development of next-generation BEVs.

Chen et al. [2] proposed a data-driven method to extract open circuit voltage (OCV) curves from large-scale battery operating data. In this method, OCV segments are online identified using a Thevenin model and then concatenated to form a complete OCV curve. A series of constraints have been imposed to ensure high reliability. The obtained OCV curves can be utilized as the basis for state estimation and health diagnosis. Therefore, this method circumvents long-term OCV tests by taking full advantage of battery operating data.

Zhou et al. [3] investigated the nonlinear degradation prediction of lithium ion batteries using a transfer learning strategy. Battery ageing features are identified and fed into a deep neural network to predict battery ageing. Then, transfer learning is implemented through the Bayesian model migration. Their results demonstrate high life prediction accuracy using only the first 30% degradation data in the presence of rollover failure. By accurately predicting battery degradation, this study can help update management studies to prolong battery life.

Li et al. [4] developed a fault diagnosis method for batteries in EVs based on signal decomposition and feature clustering. They employ symplectic geometry mode decomposition and a distance-based method to characterize the similarity of battery states. Then, a feature clustering method is proposed to detect cells with potential faults. Validations based on real-world EV data indicate that this method can effectively detect abnormal batteries 43 days ahead of the occurrence of thermal runaways. Therefore, this study can be used to avoid catastrophic battery failures.

Yu et al. [5] proposed a method to estimate and correct branch currents in battery packs. Two neural networks are designed to mitigate the cost of measuring branch currents. One neural network is adopted for current estimation and the other is used to alleviate the influence of current pulses. The proposed method is validated using multiple profiles and demonstrates superior performance. This study helps in reducing the hardware costs of onboard battery systems by reducing the number of sensors.

While most battery modeling and characterization techniques focus on electrical and thermal aspects, the mechanical performance of batteries is critical to ensure the safety of battery systems in EVs. In view of this, Li et al. [6] analyzed the mechanical structure of batteries and developed a first-principle battery mechanical model. The vibration signals during battery charging and discharging were measured through a laser Doppler vibrometer, and were utilized for model parameterization and validation. Their experimental validation results demonstrate the accuracy of the developed model and highlight the significance of mechanical models in the management of onboard battery systems.

Tian et al. [7] investigated the safety issue faced by the wireless charging of EVs. Although wireless charging has the benefit of high convenience and is promising for self-driving applications, its applications require the detection of foreign objects. In this regard, Tian et al. reviewed the state-of-the-art techniques for the detection of metal objects

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and living objects in the context of EV wireless charging. The advantages and limits of different methods are discussed, covering the aspects of sensitivity, reliability, adaptability, complexity, and costs. On this basis, future challenges and research trends are discussed, aiming to promote the commercialization of safe EV wireless charging.

Wang et al. [8] reviewed the applications of digital twin in energy sector. They analyzed the key technologies of digital twin and highlighted their various energy applications, including the low-carbon city, smart grid, electrified transportation and advanced energy storage systems. After discussions on the techniques and applications of digital twin technique, the corresponding challenges are identified with several promising solutions. This review clarifies the role of the digital twin in advancing energy systems and is promising to inspire innovations in this area.

Bidirectional DC–DC converters play an important role in new energy vehicles powered by batteries and ultracapacitors. In this regard, Wang et al. [9] reviewed the development of the topologies of the bidirectional DC–DC converters. An evaluation system was first established, based on which eight typical non-isolated converters and seven isolated converters were assessed. After comprehensive discussions in terms of the existing bidirectional DC–DC converters, the authors further analyzed the issues of existing converters and pointed out future research directions.

Xu et al. [10] investigated the energy management of hybrid EVs for enhancing energy efficiency. Deep reinforcement learning (DRL), which is a game changer in many other fields, is introduced to design energy management strategy. Furthermore, the authors propose an efficient transfer learning strategy for the generalization ability of the DRL in the presence of varying driving profiles. The authors show that the proposed strategy can be well generalized to different driving cycles with faster convergence. This study highlights the opportunities for using artificial intelligence to advance vehicle energy management.

Wang et al. [11] studied the energy management of EVs powered by fuel cell/battery/supercapacitor energy topology. The authors first identified the limitations of vehicles relying solely on fuel cells and propose the fuel cell/battery/supercapacitor energy topology as a potential solution. Then, a corresponding energy management strategy is proposed by using the DRL. This strategy accounts for three optimization objectives, including battery life, fuel cell life and hydrogen consumption. A comparison with existing energy management strategies is carried out from different aspects, and the results indicate the superiority of the proposed one. This study demonstrates the versatile applications of the DRL technique.

In addition to energy management, the capability of DRL is also demonstrated by other tasks. Fang et al. [12] proposed a power management strategy for plug-in EVs that integrates deep reinforcement learning and driving cycle reconstruction. In this strategy, the driving conditions are divided into three cases according to the speed. One soft actor-critic network is developed for each case, and the driving reconstruction method is applied to online update the algorithms. The authors validate the proposed strategy based on standard test profiles and real-world data. Results substantiate reduced costs in comparison to benchmarks, indicating the effectiveness of artificial intelligence-based power management.

Rare-earth permanent magnet (PM) motors play a critical role in EV powertrains. To address the challenges posed by limited rare earth supply, Zheng et al. [13] reviewed the primary alternatives for less-rare-earth PM motors, including the less-rare-earth PM-dominated motor and less-rare-earth PM-assisted motor. Their working principles are analyzed together with their pros and cons. Further discussions identify the key challenges in this area and promising solutions. This review provides a roadmap for devising PMs with less rare-earth dependency.

Zhang et al. [14] designed a structure for cooling the propulsion motor of solar unmanned aircraft. The proposed structure integrates the stator windows with heat pipes. A prototype is fabricated for experimental validation. Compared with the baseline motor, the designed structure reduces the temperature rise by 35% and improves efficiency by 1.5%. Besides, the proposed structure can be applied in various propulsion systems, such as EVs and solar aircraft.

Liu et al. [15] conducted simulations and experiments to analyze the internal voltage sources of permanent magnet motors with different drive topologies in these additional current subspaces. The source and characteristics of the internal voltage sources are investigated, based on which a predictive model calibration method is devised. Besides, a deadbeat current control scheme is proposed. Their results provide insightful guidance to the controller design for permanent magnet motors.

Jia et al. [16] studied the speed trajectory control strategy which aims to alleviate battery degradation. In this strategy, a battery degradation model is established and integrated with a multi-objective optimization function. Then, the optimal vehicle speed is determined and controlled through sequential quadratic programming and model predictive control. Their validations indicate that this strategy reduces battery degradation by over 10%, resulting in prolonged battery and vehicle life. This study helps in maximizing the economic and environmental values of EVs powered by lithium ion batteries.

Hong et al. [17] designed a gear shift strategy for enhancing the riding comfort and driving dynamics of EVs. The proposed strategy is compared with the prevalent Smith controller and the results unveil that the designed controller enables smooth shifting under various conditions. Besides, the impact of time delay and the interference of external disturbance can be well mitigated. The authors also systemically discussed future research opportunities to further upgrade the design.

Hang et al. [18] proposed a cooperative decision-making framework to deal with the driving conflicts faced by connected automated vehicles in the case of unsignalized roundabouts. The proposed framework accounts for the driving preferences of vehicles and the motion prediction is adopted to assist with decision-making. Hardware-in-the-loop tests are carried out to validate the proposed framework. The experimental results show that motion prediction successfully improves effectiveness and real-time performance and the driving conflicts at unsignalized roundabouts can be addressed. This study provides a significant approach to ensuring the safety and efficiency of connected automated vehicles.

Han et al. [19] investigated the driving safety of vehicles. They propose a segmented trajectory planning strategy to ensure that vehicles can smoothly avoid collisions. This strategy accounts for a variety of constraints such as the vehicle states to determine an optimal trajectory from a set of lane-change trajectories. The authors carry out simulations and experiments to demonstrate that the proposed method can efficiently avoid collisions, thereby improving the safety of vehicles.

Guo et al. [20] proposed a method to generate the scenario library for improving the safety test of driver-automation cooperation. Their method accounts for not only scenario occurrence probability but also the driver-handling challenges. A significance function is established by accounting for drivers' safety risks under different time-to-collision. Simulations demonstrate the effectiveness of this method, indicating that safety scenarios without challenges can be screened. Therefore, this study can help accelerate safety tests of vehicles.

Liu et al. [21] utilized the Stackelberg game theory to design a novel shared steering control strategy for reducing the workload of drivers and improving vehicle safety. This strategy accounts for the neuromuscular delay characteristics and can achieve theoretically optimal control. Simulations and virtual driving experiments were conducted for validation purposes. The results unveil that the proposed strategy successfully improves vehicle safety and reduces control costs. In addition, it can be used to assist unskilled drivers with better vehicle control performance.

3. Closing remarks

The research articles published in this special issue provide new insights into i) the design of battery management strategies to ensure long life and high safety of battery systems; ii) trajectory planning strategy that can avoid possible safety issues of vehicles; iii) energy management strategies with high efficiency; and iv) resource friendly EV motors with better performance. In addition, the emerging artificial intelligence techniques such as machine learning [3,5,10–12] and digital twin [8] demonstrate promising applicability in various aspects of EV research. The findings and highlighted future research trends will help the community improve EV design and control and are expected to stimulate further contributions to the development of EVs.

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